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CHANGES OF FERROMAGNETIC MINERALS AND THEIR BEARING ON MAGNETIC PROPERTIES OF ROCKS

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ABSTRACT

Physical and chemical properties of samples from a group of pre-Cambrian diabase dikes characterized by an inverse magnetization have been determined. There are present two different ferromagnetic minerals intimately intermingled with one another and with non-magnetic ilmenite. Experiments to duplicate the original thermomagnetic behavior of these rocks cannot be carried out because by heating, non-reversible changes of physical structure are produced which alter the magnetic properties of the sample. Magnetization experiments show that the inverse magnetization is produced by particles having high coercive force; in natural samples a large part of the ferromagnetic material is demagnetized but can be easily magnetized. The hypothesis is advanced that the inverse magnetizations of these dikes having uniform magnetization could have been produced with the earth's magnetic field in its present-day sense, as a result of the partial low-temperature oxidation of primary magnetite forming small particles of high coercive force in the reversed induction field adjacent to large magnetite particles and the subsequent demagnetization of some of the larger particles. Various implications of the hypothesis are pointed out. The gross scattering of directions of magnetization observed in two dikes are proved not to have been caused by turbulent movements below the Curie point; the scatter is attributed to chemical changes, but no detailed mechanism for producing the scatter is proposed.

I—INTRODUCTION

In an earlier note [see 1 of "References" at end of paper] some speculations were made about a possible relationship of chemical composition and the magnetic properties of igneous rocks. The author's interest in this topic was heightened by the observations reported recently by Balsley [2] and by Balsley, Buddington, and Fahey [3] of an apparent correlation of composition with the magnetizations of igneous rocks. Briefly, in our note it was suggested that the inverse polarizations of certain igneous rocks might have been acquired even with the earth's field in its normal sense, provided that the composition of the ferromagnetic ingredient present was favorable to exsolution processes taking place, and that the rate of

cooling of the melt was slow enough that interacting magnetic phases could form This paper reports the results of an examination of that suggestion, and of other related topics.

Initially it was the aim of the study to get some insight into the relationship of exsolution phenomena and magnetic properties of rocks by careful determination of the dependence of magnetic properties on the position of the sample relative to the chilled margins of the intrusive. It was anticipated that normal polarizations would be observed at the margins if the ferromagnetic ingredient should cool there as a single metastable phase, whereas inverse polarizations ought to be observed in the center of the same intrusive where slower cooling would favor the development of exsolution textures.

Field studies in diabase dikes of various widths, having pronounced inverse magnetizations, did not bring out any such relationships [4], but from subsequent work in the laboratory, various interesting properties bearing on the magnetic behavior of the material in the dikes have been uncovered. To the extent that these rocks and their ferromagnetic ingredients are typical of many others, the observations and the properties described are of importance in the problem of understanding the origin and significance of magnetism in rocks.

II—FIELD OBSERVATIONS

For the field work which took place during June and July of 1952, an extensive group of dikes in Baraga County, Michigan, was selected. Balsley, James, and Wier had previously described [5] the striking aeromagnetic anomalies produced by these dikes and had demonstrated that representative samples have a magnetization that is approximately opposite in sense to the earth's present field. They generously made available additional data on the dikes that were not included in their report.

Figure 1 indicates some of the regional development of the dike swarm as determined by the airborne magnetometer, and shows localities where samples could be obtained. The area generally is one of extensive glacial cover, marsh, and dense second growth, and from the rare outcrops the extent of the dikes is not obvious.

All of the dikes sampled are remarkably well-preserved, and to the naked eye only rarely show signs of internal weathering and oxidation of the normally grey-blue diabase. Contacts with the country rocks, when found, are sharp, and show a conspicuous grain-size gradation that indicates chilling by the borders. No evidence of assimilation of the country rock by the diabase was found. There is little reason to doubt that four of the five dikes sampled are comagmatic; the fifth dike (Herman 42′) may date to another epoch of igneous activity, for its specific moment is significantly smaller and the strike of its walls does not conform to the regional strike of the inverse swarm of which the other dikes apparently are members. From their areal development in metamorphosed Huronian sediments and their truncation by flat-lying Cambrian sediments, these dikes are believed [6] to be late pre-Cambrian in age. No evidence was found that these dikes have been deformed or disturbed since the time of their emplacement.

Oriented samples for magnetic measurement were cut out by two methods:

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by coring *in situ* with a portable rig that is powered by a generator in the field truck, and in the case of the samples from the Covington 150′ dike, by removing oriented blocks and coring them on a drill-press in the truck. Cores are 1″ in diameter and were cut to an inch length by a single pass through a properly-spaced pair of diamond saws. Water cooling was used on all the cutting operations.

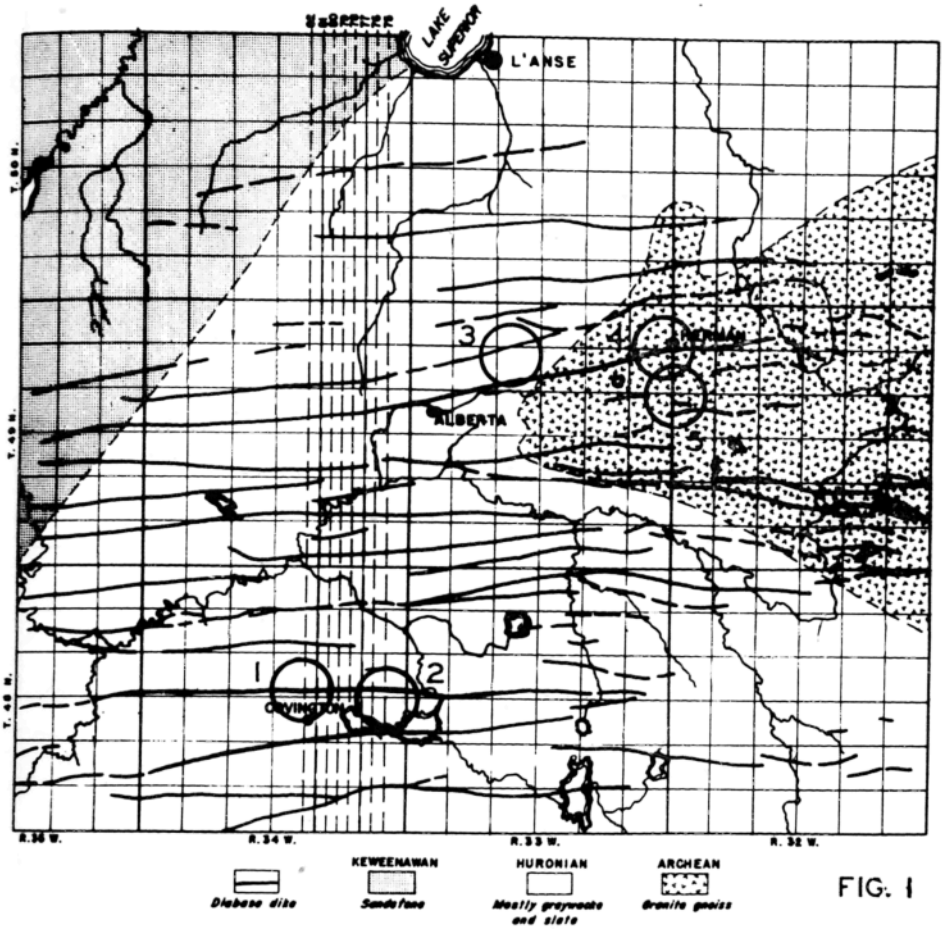


FIG. 1

1=COVINGTON 150′ 2=COVINGTON 1.6′ 3=TAYLOR MINE 40′ 4=HERMAN 42′ 5=HERMAN 5′
(MAP AND AEROMAGNETIC DATA BY COURTESY OF BALSLEY, JAMES, AND WIER, U.S. GEOL. SUR.)

Measurements of the remanent moments were made while in the neighborhood of the outcrops with the "spinner magnetometer," a more portable and more sensitive version of the instrument [7] used previously. In the case of these strongly polarized rocks, with the sampling and plotting techniques employed, the estimated probable error for the plotted directions of magnetization is 4°. The plots are exclusively in the lower hemisphere of the Schmidt equal-area projection.

Three of the five dikes have uniformly the same directions of magnetization;

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data from two of them and the sampled locations relative to the enclosing contacts are indicated in Figures 2*A* through 2*D*. The data from the Covington 150′ dike were few owing to sampling difficulties and are not presented. They were sufficient, nevertheless, to confirm the fact that this dike is responsible for one of the large

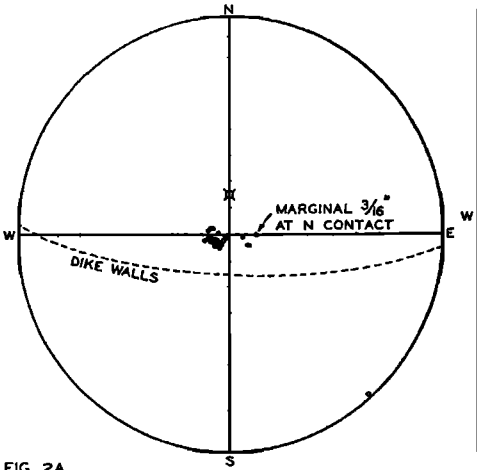
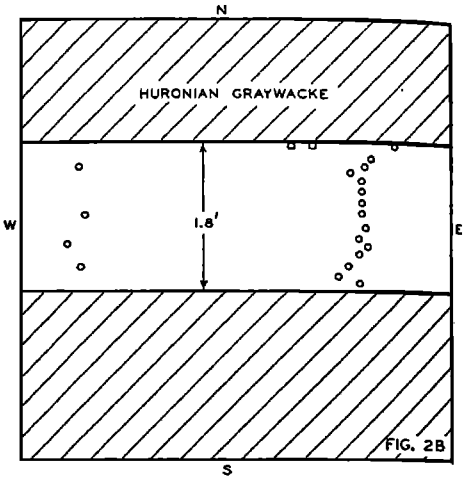


FIG. 2A
DIRECTIONS OF MAGNETIZATION, COVINGTON 1.8′ DIKE
•=S SEEKING POLARIZATIONS
⌘=EARTH'S PRESENT FIELD



PLAN OF COVINGTON 1.8′ DIKE SHOWING SAMPLE LOCATIONS

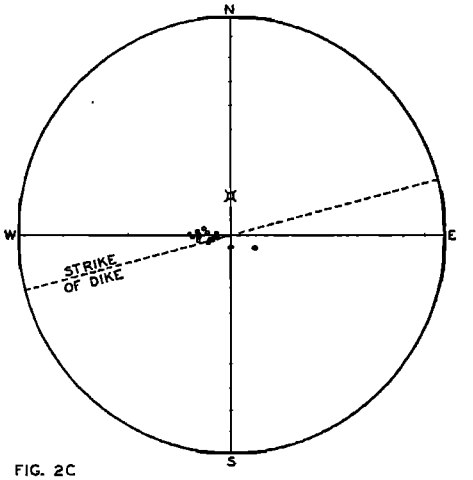
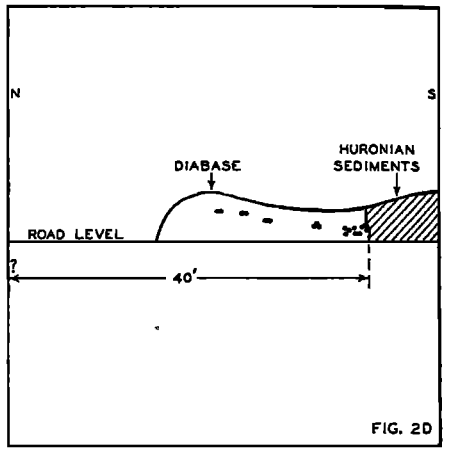


FIG. 2C
DIRECTIONS OF MAGNETIZATION TAYLOR MINE 40′ DIKE
•=S SEEKING POLARIZATIONS
⌘=EARTH'S PRESENT FIELD



ELEVATION OF TAYLOR MINE 40′ DIKE SHOWING SAMPLE LOCATIONS

observed aeromagnetic anomalies and to show the correlation with the two smaller dikes having the same magnetization.

Two other dikes have magnetizations that do not conform at all to this picture of uniformity; findings in them are discussed in a later section.

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Interpretation of field observations

It has generally been customary to interpret uniform directions of magnetization of igneous rocks as indicating the actual sense and direction of the earth's magnetic field present at the time the rocks cooled from above their Curie point. This interpretation has been criticized [1] and the proposal made that magnetizations opposed to the applied field could arise under appropriate circumstances where two intimately associated magnetic phases are present during cooling. We now have evidence that changes leading to reversal may occur in the two intimate magnetic phases long after cooling.

In the present study, sampling has been carried out in a way that should show whether or not quenching has any marked influence on the sense of the direction of magnetization of the igneous body. From the field observations alone, one must believe that such quenching as did occur on the margins of the dikes had no influence on the sense of the magnetization observed today in the diabase. The dike material is uniformly magnetized to within at least the outer 1/8″ of the Covington 1.8′ dike, for example, and there is no support here of the suggestion that quenching favored the development of a single metastable phase with normal polarizations in a dike which is generally characterized by an inverse magnetization. These observations hence give no support to Néel's [8] treatment calling for two phases; likewise, they are not necessarily in conflict with his proposals, for the influence of quenching was a matter of conjecture in the first place.

III—LABORATORY INVESTIGATIONS

Heating experiments

The first experiment on this dike material was to determine the nature of the magnetization produced when specimens are allowed to cool in the earth's present magnetic field to room temperature from above their Curie point. When representative samples of all these dikes are heated to 600°C and allowed to cool in the earth's field over a period of an hour, normal polarizations are acquired that have the same sense and direction as the applied field. The new intensity of magnetization is about 10 times the intensity possessed by the natural samples. The heating was over a period of about two hours in a small electric A.C. furnace; the brass crucible that contained the samples embedded in fine quartz-sand was cooled outside the furnace.

However, the time interval of significant cooling of the 1.8′ dike may be estimated to be of the order of a few days, and on the supposition that exsolution processes might require at least this period, representative samples, protected somewhat from atmospheric oxidation by being encased in Pyrex tubes having openings through fine capillaries, were cooled gradually over a period of a week from 600°C to 100°C in the earth's field. As before, normal polarizations were obtained.

These two experiments demonstrate that at the present time this material has the inherent property of acquiring a magnetization showing the same sense and direction as the field applied during cooling. That the rock has this property today, however, is not an adequate basis for concluding that at the time of the

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initial injection, the earth's magnetic field was reversed. First it must be demonstrated that by the heating experiment no physical or chemical recombinations or redistributions of the component parts are made to take place.

X-ray diffraction studies

This question about recombinations induced by the heating experiments has been investigated by means of X-ray diffraction spectra of untreated and heat-treated portions of the fraction that could be separated magnetically from the coring-bit cuttings of the Covington 150′ dike material. I am grateful to Mr. Gordon Davis, of the Geophysical Laboratory, Carnegie Institution of Washington, for preparing the spectrograms. They were made with a "Norelco" Wide Range X-ray Spectrometer, with filtered Co Kα radiation.

For the heat treatment, approximately a gram of the magnetic fraction was placed at the bottom end of a sealed-off piece of Pyrex tubing, the other end of which was then connected to a mechanical vacuum-pump. No measurements were made of the residual pressure of the system, but judging from other experience with the pump, there must have been very little air present to allow oxidation of the sample. Pumping was continuous during the heating to 610°C in an electric A.C. furnace and the subsequent cooling outside the furnace. The whole cycle of heating and cooling required an hour. No reactions of the material with the glass were noted, but the powdered sample showed throughout its volume a conspicuous change from lusterous black to a deep ruby-red color.

From the spectrograms (Fig. 3) it can be determined that the structural arrange-

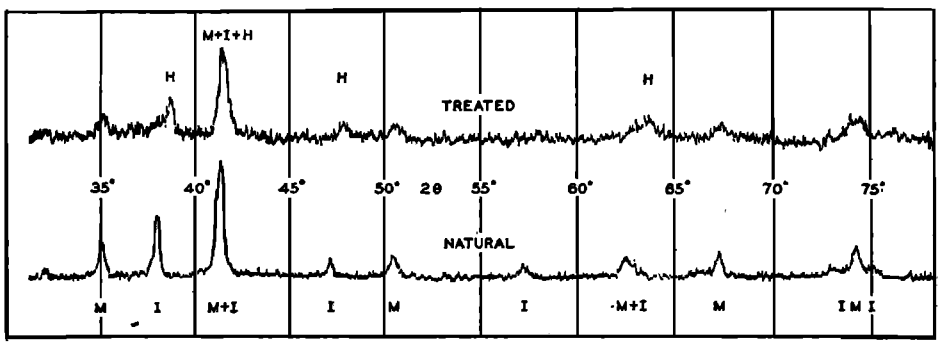


FIG. 3—TRACING OF X-RAY SPECTROGRAMS OF MAGNETIC FRACTION

ment of the material has been altered by the heat treatment. The peaks from the untreated sample are identified as those from ilmenite (I) and from some compound (M), such as magnetite, with a magnetite structure, or maghemite (γFe_2O_3). This X-ray technique without accompanying chemical analyses is not definitive for distinguishing pure magnetite from magnetite in which considerable ionic substitution has taken place, or from maghemite. The intensities of the peaks in the heat-treated material are different from those in the original, and there are in addition at least three new lines which are identified as coming from hematite (H). Lines from ilmenite show a marked decrease in intensity. The angular widths of some

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peaks are greater; this may be interpreted as resulting either from a reduction of the grain size, or from distortions of the structure by the heat treatment, or from the development of new lines. A small shift in the spacing of some of the lattice planes can be detected.

From the foregoing simple experiment it is concluded that it is not possible to conduct heating experiments that duplicate accurately the original magnetic behavior of these rocks from the time they were first injected to the time they cooled to room temperature. By the heating experiment, physical and chemical changes are produced which undoubtedly have a bearing on the magnetic properties. One of the most important changes is from the non-reversible reaction which produces hematite (essentially paramagnetic) from maghemite (strongly ferromagnetic) [9].

Coercive force spectrum

Inasmuch as the magnetic material of these dikes is of such a nature that magnetic data derived from heating experiments are of questionable significance, efforts were made to find other ways of establishing intrinsic magnetic properties. The response of the samples to steady and to alternating magnetic fields has been found of interest.

The present measurements were an outgrowth of earlier experience with the behavior of some Silurian sedimentary rocks having inverse polarizations. From field observations it had been established [10] that these sediments had retained stable polarizations for at least 200 million years, and it seemed that with them it should be possible to establish an upper limit for the intensity of the earth's magnetic field for this interval by determining the threshold limit of magnetic field that had to be applied momentarily in order to change the directions of magnetization of the samples. It was observed that a field of 15 oersteds applied at right-angles to the polarization would cause it to turn perceptibly (10°); subsequently, however, on repeat observations of these remagnetized samples, it was found that they had recovered their original directions of magnetization (±2°) during storage in the laboratory. This behavior suggested that the samples contained a mixture of magnetically "hard" and "soft" components.

The coercive force of a sample can be considered a measure of the ease with which its state of permanent magnetization can be changed, and may be defined by the intensity of the steady magnetic field that must be applied momentarily in opposition to the remanent polarization in order to bring it to zero intensity. The value of the coercive force may be determined by successive measurements following the application of fields of increased value. If exactly the correct intensity of field is applied precisely in opposition to the remanent moment, a "demagnetization" will have been effected, and the sample will have no external residual polarization.

Another method of demagnetization, commonly exploited by watchmaters, for example, is to place the sample in the field of a coil that is energized with alternating current and then slowly reduce the current to zero or slowly withdraw the sample from the coil. The explanation of these demagnetization processes in the light of what is known of the domain behavior of ferromagnetic materials is complex

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and has been treated by Williams and Goertz [11] and by Bates and Martin [12].

It has been found that these Michigan dike rocks are easy to "demagnetize" with application of a steady field in opposition to the polarization, but relatively much more difficult to demagnetize with alternating fields, and from these experiments it seems fair to say that the ferromagnetic ingredients consist of a mixture of parts, having on the one extreme a low coercive force, and on the other a high coercive force.

For example, Figures 4*A* and 4*B* show that sample 1212 from within 5″ of the wall of the Covington 1.8′ dike can be "demagnetized" with momentary application of a steady field of 95 oersteds; with a field of 110 oersteds, the direction of the remanent moment has been approximately reversed and has an intensity of about 20 per cent of the original value. If, now, this reversed sample is subjected to successively greater increments of alternating fields, it is observed that the remanent moment first decreases through zero, then increases in intensity in reversed direction (now approaching the original direction), and when alternating fields of 150 oersteds peak intensity are reached, the sample has recovered 97 per cent of its original intensity of magnetization. During the applications of the magnetic fields the sample was held each time as closely as possible with its previously determined axis of magnetization parallel to the direction of the applied field. Slight non-parallelism is believed to account for the break in the observed changes of direction and intensity.

A simple interpretation of these observations is that the natural remanent moment of the sample is accounted for by the presence of high coercive-force material having inverse magnetization. Also present is a significant amount of essentially demagnetized material which contributes little to the external magnetic field of the original sample.

From other similar experiments the generalizations can be made that the "soft" components of the rocks are not magnetized appreciably (< 10 per cent), and there is a dependence of coercive force on position of the sample relative to the dike walls, the finer grained samples having somewhat higher coercive force.

Significance of high coercive force

The subject of coercive force of ferromagnetic materials has received intensive treatment in recent years (see, for examples, [13], [14], and [15]) in connection with the properties of magnetic materials and with the problems of the magnetization mechanism of domains. Many factors influence coercive force, but grain size may be one of the most important, for when a particle is smaller than a certain dimension characteristic of the material, it becomes impossible to form, within the grain, a domain boundary wall and the particle throughout is magnetized to saturation. In order to reverse the direction of magnetization of one of these small particles, all the polarized electron spins contributing to the magnetic moment must be forcibly reversed. To do this requires greater field strength than is required to translate a domain boundary within a larger grain.

Galt [16] has demonstrated that the coercive force of a pure synthetic magnetite single crystal of high perfection is 0.1 oersted. In contrast, the coercive force to alternating fields of a sample from the center of the Covington 150′ dike, where

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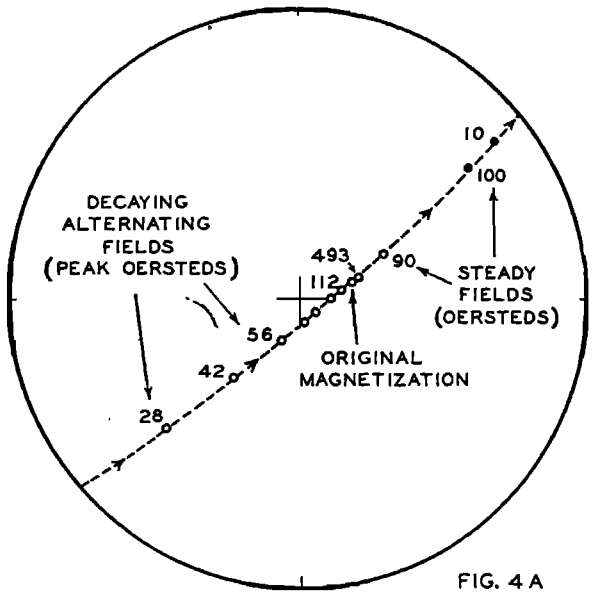


FIG. 4 A

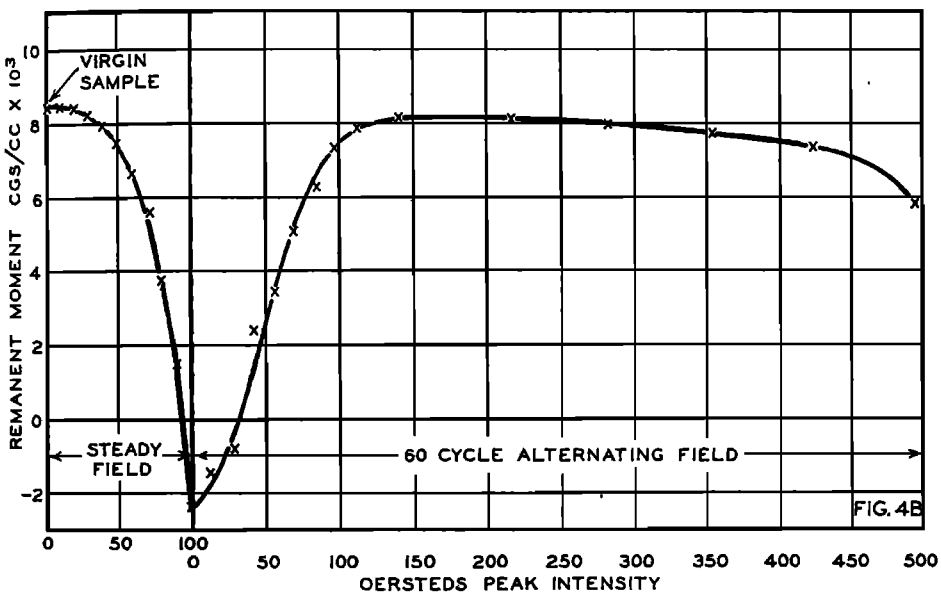


FIG. 4—CHANGES OF (A) DIRECTION AND (B) INTENSITY OF MAGNETIZATION PRODUCED BY STEADY AND ALTERNATING MAGNETIC FIELDS; COVINGTON 1.8′ DIKE SAMPLE

the "magnetite" grains measure up to 3 mm across (comparable to the size of Galt's crystal), is something over 200 oersteds.

Some of this ferromagnetic material was examined under the metallographic microscope in an effort to account for the high coercive force. I am grateful to Dr. J. W. Greig, of the Geophysical Laboratory, Carnegie Institution of Washing-

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ton, for preparing the polished specimens, and for giving much valuable information.

At least three different phases are present in this material, and the size of scale on which individual phases can be distinguished ranges from a small fraction of a millimeter down to, and presumably below, the limits of resolution of the optical microscope (0.2 μ). Boundaries between phases commonly appear curved and irregular, and frequently grains are seen having a reticulation that is characteristically developed in magnetite from which excess titanium has exsolved in the form of crystallographically-oriented ilmenite [17]. A comprehensive study of these textures was not made, but the examination was sufficient to justify the belief that the high coercive force results from the minute subdivision of the bulk grain into an unknown large number of much smaller individual particles.

A further demonstration was made of the minute scale on which the ferromagnetic material is subdivided into magnetic and non-magnetic fractions. A quantity of the magnetic fraction, about the volume of a grain of rice, was slightly moistened and ground in an agate mortar for a few minutes. This powder was then suspended in a few cubic centimeters of water. It was sufficiently finely divided that much of it remained in suspension for many minutes. However, with a bar magnet it was possible to attract practically all of the suspension to the wall of the container. A similar test with ilmenite showed that this mineral, ground to corresponding size, cannot be removed from the suspension with the same magnetic separation technique. These tests imply that even with very fine grinding, particles of ilmenite are not broken away pure from the magnetic material, and hence the size scale on which the ilmenite is developed is small.

IV—HYPOTHESIS ON THE INVERSE MAGNETIZATION MECHANISM

Our knowledge of the mechanism by which these Michigan dikes became magnetized must be derived by means less direct than a heating experiment, inasmuch as original structures are modified by the experiment itself which otherwise might be decisive.

A hypothesis is now proposed for the magnetization mechanism which depends basically on the following factors:

(1) The earth's field at the time of cooling of the dikes was in the same sense and approximate direction as today.
(2) The stable magnetic moment with high coercive force results from the presence of an intimate mixture on a very small scale of maghemite and some ferromagnetic ferrite, presumably magnetite.
(3) A significant portion of the ferromagnetic material present is not contributing to the external natural remanent moment.
(4) The ferromagnetic ingredients have intimately mixed in with them a significant quantity of non-ferromagnetic foreign material (ilmenite).

The first factor is the unknown variable we are particularly interested in understanding; the other factors have been demonstrated true by observation.

The hypothesis follows. At high temperatures, magnetite can contain within

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it appreciable quantities of titanium in homogeneous solid solution, but at some unknown lower temperature, titanium in excess of some percentage which has not yet been established, becomes unwanted by the structure and is rejected as ilmenite ($FeTiO_3$) in the form of irregular volumes and exsolution lamellae parallel to the {111} of magnetite (for illustrations and discussion of numerous examples, see [17]). By comparison with magnetite, ilmenite can be considered non-magnetic. Hence, by this exsolution process, a high-temperature homogeneous grain of "magnetite" on cooling is subdivided into a myriad of slabs or zones of somewhat impure magnetite separated from one another by smaller volumes of non-magnetic ilmenite. The size scale of exsolution textures probably depends on the time-temperature relations during cooling as well as on composition. It is not known whether the exsolution process begins to take place above or below the Curie point of the magnetite. All ferromagnetic parts of these complex grains, by the well-known thermoremanent magnetization process, become magnetized to high intensity in the normal sense on cooling in the earth's field.* Maghemite is not a primary constituent of igneous rocks, but can be created [18, 19] within them by the low temperature oxidation of magnetite, perhaps as the result of the permeation of late-stage hydrothermal solutions, or by other deuteric reactions. It may be anticipated that the smaller particles of magnetite within the complex grain, owing to their higher surface energy, will be preferentially converted to maghemite (strongly ferromagnetic [9]). During their composition transition, the magnetizations of the particles undergoing alteration will be unstable and the magnetic field imposed on them by their environment will determine their new direction and sense of magnetization.

The problem of the sense and intensity of the external magnetic field of a sample having two magnetic components whose fields interact with one another has been treated by Néel [8]. He has derived a set of conditions that will allow the sense of magnetization of the sample to reverse as a result of the fields of secondarily magnetized material externally overbalancing the fields produced by material initially magnetized in the normal sense. Among other conditions, his treatment calls for two components with different Curie points, the external fields of the first magnetized component A being responsible for imposing an inverse magnetization on the material B with the lower Curie point during the cooling process. Alternatively, he has derived the conditions which will allow a reversal of the moment to take place at low temperature if by selective chemical action the first magnetized component A is eliminated.

In the present hypothesis, certain modifications are made to these proposals in order to take into account the facts of observation. It is suggested that initially there needs to be but one constituent A (magnetite) dispersed by the non-magnetic ilmenite; late-stage oxidation reactions selectively create B (maghemite) at the expense of the smaller particles of A, and it is magnetized in the inverse sense by the external fields of the first magnetized component A in the manner treated by Néel. The external moment of the sample then later becomes reversed as a result

*The possibility is not considered here that exsolution textures reduce the macroscopic particles to smaller individual volumes in the size range where there is a dependence of Curie point on dimensions.

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of the demagnetization, in time, of the larger particles of component A. Magnetite grains that are too large to remain as domains for long periods can be expected to demagnetize by forming domain field closure loops within themselves. The smaller size volumes of maghemite with the inverse moment will remain as stable domains with high coercive force. The essential rôle of the titanium impurity is that it interrupts the individual large grains so that they can behave as an aggregate of independent tiny particles whose fields interact. Similar conditions might equally well have been achieved by impurities other than titanium coming out of high-temperature stable solid solution in the magnetite.

Criticism of the hypothesis

It is believed that the elements of a reasonable working hypothesis are here advanced for explaining how these particular dikes might have acquired their inverse magnetizations without requiring that at one time the earth's magnetic field was reversed. The hypothesis is consistent with the facts of observation presented, but cannot be considered supported in detail because of the present-day lack of knowledge of the evolution of composition of "magnetite" crystallizing initially from a melt of basaltic composition from high temperature and reacting with residual liquids down through low temperatures. Furthermore, the information available in the literature indicates that the properties of maghemite are so variable that quoted values of its "constants" cannot be applied to these rocks without specific confirmation. Gheith's studies [20] indicate that the transition temperature of maghemite to hematite depends on the mode and temperature of formation and environment of the sample, and varies from something under 450°C to 568°C. Néel states [21, p. 250] that the transition takes place between 400°C and 600°C. This variability has been interpreted [22, 23] as resulting from partial substitution of various divalent ions for ferrous ions of the magnetite which has been oxidized. Forrer [24] has mentioned various magnetic and crystallographic properties of ferric oxides produced by the oxidation and dehydration of a ferrous oxide hydrate. No information could be found about maghemite which would appear to invalidate the hypothesis, and the hypothesis seems consistent with the facts available.

Implications of the hypothesis

This hypothesis suggests that under appropriate conditions of composition and chemical history, it should be possible for an igneous rock to acquire an inverse magnetization with the earth's field retaining its present-day sense. Rocks recognized as having an inverse magnetization generally attract attention because the polarizations call for an explanation on the basis of either one of two interesting hypotheses: reversal of the earth's field, or a self-reversal of the remanent moment. Normal polarizations, requiring no special explanation, are somewhat taken for granted, and do not receive the same critical inspection. It would be of considerable interest to know how the properties of rocks having normal and inverse polarizations compare when subjected to the same tests. Indeed, this study must accompany any claim that an established self-reversal mechanism is proof of a constant earth's field.

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As a part of this hypothesis it can be imagined possible, in other cases, that during the end stages of the cooling of a melt of appropriate composition, there may be no appreciable reactions to produce maghemite within the ferromagnetic minerals containing exsolution textures. The igneous body would then have a normal polarization at the completion of the cooling. If this material were now to be eroded away and deposited as a sediment, it would have latent within it the property of self-reversal. Subsequent to deposition of the sediment and its magnetization in the normal sense by the orientation during settling of the ferromagnetic grains, an appropriate temporary change of chemical environment (percolation of heated solutions, etc.) could promote the partial oxidation of the magnetite, and with the demagnetization of the remainder, the self-reversal would take place. The sediment would thus have an inverse moment where the alteration had been effective. Transitions between the two extremes would be represented by samples having relatively small magnetic moments but comparable susceptibilities.

A process of this general character appears to have taken place in some Silurian sediments of the Appalachian geosyncline, but not in sediments outside the geosyncline [25]. Other geologic settings for such a process can be imagined. If a sediment possessing a latent self-reversal property were to be invaded by a dike, for example, it might show an inverse polarization only in the vicinity of the dike where oxidation reactions at higher temperatures are promoted; the dike itself would have normal or inverse magnetization, depending on its composition and history. If the dike had an inverse polarization, the tentative conclusions would be reasonable, though inadequately founded, that the earth's magnetic field was reversed at the time the dike invaded the sediments and that they acquired their local inverse polarization by the normal thermoremanent magnetization process in a reversed earth's field. These possibilities might be profitably explored by comparisons of the polarizations of adjacent beds having different solution permeabilities; another interesting possibility is that by appropriate chemical manipulation it might be possible synthetically to induce a reversal of the moment of the sediment.

V—DISCUSSION

In the foregoing experiments, a number of procedures have been employed for defining properties that are pertinent to understanding the magnetization mechanism of these rocks. Particular emphasis is given to the fact that with these samples it is not possible to perform heating experiments which duplicate the original thermomagnetic behavior owing, first, to the presence of unstable secondary ferromagnetic maghemite that was produced by low temperature oxidation of magnetite, and, second, owing to the fact that ilmenite, present originally, is resorbed by the magnetite at the temperatures used in the heating experiment and is retained, presumably as a metastable solid solution, when the specimen is recooled to room temperature. These two factors make it impossible by direct means to demonstrate the mechanism by which these rocks became magnetized.

Many data are available which suggest that comparable changes of structure take place in some other rocks during heating experiments. It has been known for some time [26] and recently demonstrated [27, 28] that the magnetic suscepti-

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bility *versus* temperature curves obtained in the initial heating of common rocks are frequently not duplicated by measurements made during the cooling part of the cycle and on subsequent runs. This behavior is especially pronounced in a maghemite-rich gabbro studied by Nagata [29].

Unless evidence to the contrary is presented, it seems prudent to anticipate that alterations and recombinations comparable to those described here are likely to take place in many rocks; therefore, conclusions based exclusively on data derived from heating experiments are of doubtful or negative value in the problem of describing the history of the earth's magnetic field. To date it does not seem that any claim of a reversal of the earth's magnetic field has been adequately substantiated or justified.

VI—"TURBULENT" DIKES

Two other dikes in the same area have a pattern of polarizations that is in marked contrast to the uniformity shown by the dikes that have just been discussed. The plots of the observed polarizations (Figs. 5 and 6) suggest gross move-

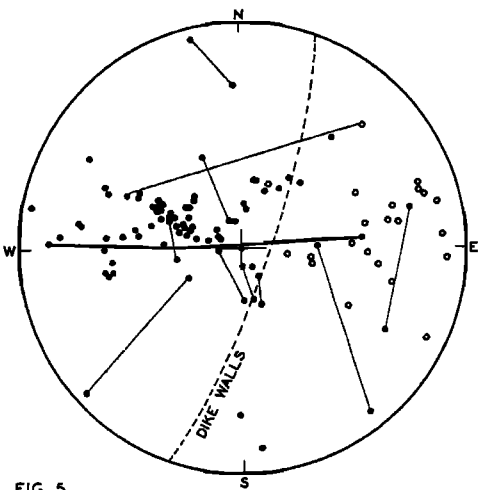


FIG. 5
DIRECTIONS OF MAGNETIZATION, HERMAN 42' DIKE
• N SEEKING POLARIZATIONS
○ S SEEKING POLARIZATIONS
⊷ SAMPLES ≤ 12" APART

ments of the dike material below the Curie point. Similar observations have been made and treated from the standpoint of turbulence by Bruckshaw and Robertson [30]. The Michigan dikes have been examined for evidence of turbulence, and evidence to the contrary has been found.

Informal discussions with a number of petrologists have brought out the general opinion that diabase in a dike is not likely to behave like a liquid below 900°C for a probable low limit, and 800°C for a rather certain low limit. Since the Curie point of this material is below 600°C, in order to produce a "turbulent" pattern in the directions of magnetization, it was considered more satisfactory to imagine that the "turbulence" resulted from multiple injection processes: a dike

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is formed, solidified, magnetized, broken up perhaps by renewed movements of the dike walls or cooling cracks, and then the fractured structure is reinvaded by molten diabase, at which time the directions of magnetization of the original material are scattered at random.

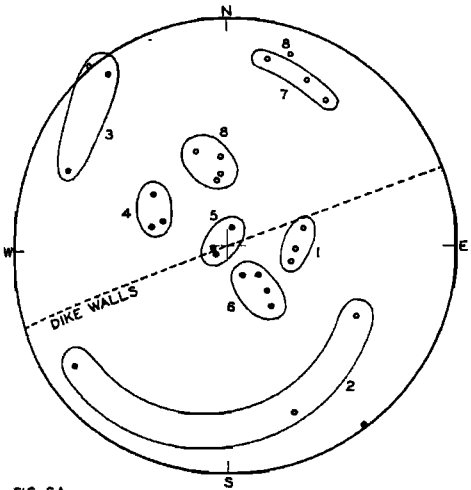
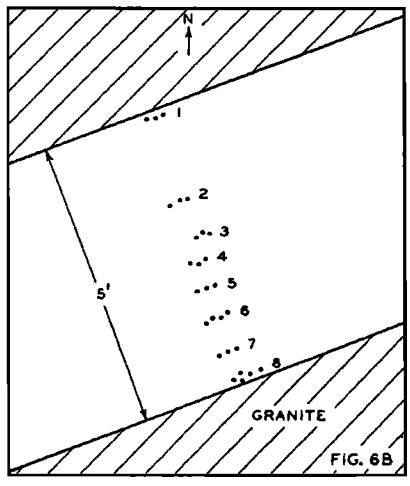


FIG. 6A
DIRECTIONS OF MAGNETIZATION, HERMAN 5′ DIKE
• N SEEKING POLARIZATIONS
∘ S SEEKING POLARIZATIONS



PLAN OF HERMAN 5′ DIKE SHOWING SAMPLE LOCATIONS
(NUMBERED GROUPS ON FIG. 6A AND 6B CORRESPOND)

Although field evidence for such events could not be found, it was decided to put the unattractive interpretations to another test. It is well known [31] that in any flowing liquid there will be dimensional orientation of non-spherical rigid bodies contained in the liquid when there are velocity gradients of the sort that would be produced by the dike walls. It was supposed that in the dikes having uniform polarizations there should be a systematic arrangement of the long-dimension axes of the minerals formed at high temperature before injection, whereas in the "turbulent" dikes there should be no corresponding simplicity.

Intended originally for other purposes, we had constructed an instrument for measuring in any selected direction the magnetic susceptibility of rock samples. By a systematic procedure, the principal axes of the surface describing the susceptibility can be easily located, with an uncertainty, in the case of the present material, of only a few degrees. Inasmuch as susceptibility anisotropy is related to dimensional elongation of the ferromagnetic minerals, it seemed that with this equipment it should be directly possible to establish the character of the turbulent flow in these dikes. Fuller discussion of this equipment and of the significance of susceptibility anisotropy in rocks will be published elsewhere.

The results of the measurements are indicated in Figure 7. It will be seen that from place to place in the Herman 5′ dike having "turbulent" polarizations, there is a uniformity of orientation of the ellipsoid by which the susceptibility anisotropy may be described. For comparison, the corresponding data are given (Fig. 7*B*)

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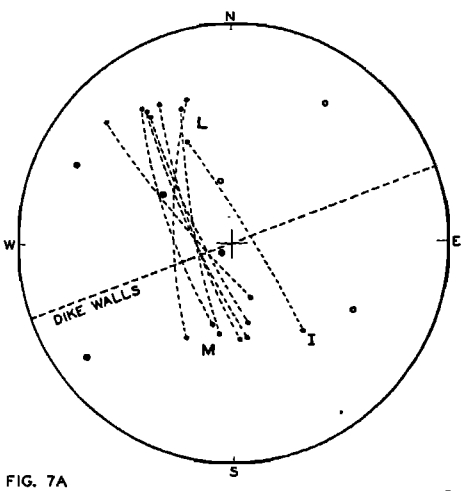


FIG. 7A
REMANENT MAGNETIZATIONS AND SUSCEPTIBILITY AXES, HERMAN 5′ DIKE

●=N SEEKING POLARIZATIONS
○=S SEEKING POLARIZATIONS
←→=ASSOCIATED PRINCIPAL AXES OF SUSCEPTIBILITY
L=LEAST SUSCEPTIBILITY AXIS
I=INTERMEDIATE SUSCEPTIBILITY AXIS
M=MAXIMUM SUSCEPTIBILITY AXIS

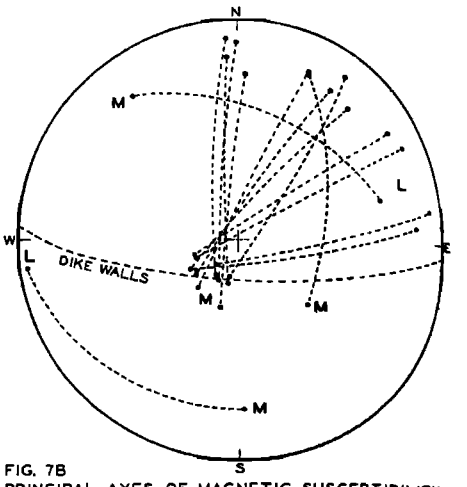


FIG. 7B
PRINCIPAL AXES OF MAGNETIC SUSCEPTIBILITY, COVINGTON 1.8′ DIKE

←→=ASSOCIATED PRINCIPAL AXES
L=LEAST SUSCEPTIBILITY AXIS
M=MAXIMUM SUSCEPTIBILITY AXIS

for the Covington 1.8′ dike. Inasmuch as these dikes have not been overprinted by a deformation subsequent to their injection, we can conclude that these patterns of susceptibility anisotropy date to, and in some way are related to, the primary injection flow of the diabase. The observed uniformity rules out turbulent flow or fracturing and reinvasion of the dike below the Curie point as acceptable interpretations of these "turbulent" polarizations.

While at the outcrops, efforts were made to recognize other mechanisms by which the "turbulent" polarizations might be explained. Owing to the intense magnetization of the Herman 5′ dike, it was possible to get an approximate idea of its state of magnetization in many places, in addition to those sampled, by means of a dip needle made with an Alnico V bar magnet measuring about $1 \times 1 \times 20$ mm supported in jeweled trunnions. With this instrument, the external field produced by the dike could be rapidly explored in some detail. It was found that the magnetic field of the dike exceeded the earth's field at a distance as much as a few inches from the dike surface. As the instrument was translated along the dike, the needle responded vigorously to changes in the direction of the external field. Approximately 100 square feet of the dike were available for study by this technique, and no simple interpretation could be made about causes for the changes in direction of magnetization from point to point. Remagnetization by lightning bolts seems to be ruled out by the fact that reversals take place in as little as an inch on smooth, homogeneous, overhanging surfaces. Explanation on the basis of elastic stresses seems to be ruled out by the fact that there is no obvious correlation of change in external field with the jointing of the rock or with lack of jointing. There is some visible evidence of variation of chemical composition of the diabase in some places, particularly along joints, but slightly bleached and altered sections appear to have the same variability of magnetization as homogeneous surfaces.

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Significance of "turbulent" magnetizations

By the susceptibility anisotropy measurements it has been demonstrated that magmatic movements below the Curie point cannot account satisfactorily for the scattered directions of magnetization of the Herman 5′ dike. Other simple interpretations seem likewise unsatisfactory. Since the dike is continuous and essentially homogeneous in its content of ferromagnetic minerals, it is reasonable to believe that at one time it was magnetized essentially uniformly in direction throughout. Because this dike appears to be closely related to the uniform inverse dikes whose magnetization has been interpreted as resulting from a dynamic process, it is appropriate to wonder whether the scattered directions of magnetization should be regarded as representing intermediate steps in a reversal process. The relationship reported by Balsley [2] of a correlation of the inclination of the magnetization with the chemical composition of the sample is perhaps pertinent to this problem; further studies should be rewarding.

VII—ACKNOWLEDGMENTS

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